

Research interests: Diffusion models, score-based generative modeling, flow-based generative models, LLMs, implicit biases in neural networks, AI interpretability, security, and safety.

Professional Experience

- Present **AI Security Evaluation Specialist**, ANSSI – French National Cybersecurity Agency
- Contributing to a roadmap for improving techniques to evaluate the security of AI systems, within the agency's newly formed AI unit.
 - Surveying, testing, and improving state-of-the-art AI security evaluation methods, in coordination with INESIA and international partners.
 - Building a strategic view of the security impacts of emerging AI uses (including automation), and scoping internal and collaborative research projects.
- Feb. 2026 – **SPAR Fellow**, Kairos, SPARAI
- May 2026 ○ Bridging RAG and RL for Safety-Critical Systems (In-Context Learning and Test Time Learning, GRPO).
- Feb. 2025 – **ML Engineer**, INRAE – National Research Institute for Agriculture, Food and the Environment, Biostatistics & Spatial Processes Unit
- Jul. 2026
- Developed transformer-based diffusion models for probabilistic imputation and forecasting of hydrological time series at the Cigéo nuclear waste repository.
 - Brought the codebase to production quality and shipped it to the ANDRA hydrology team, adapting to operational constraints and user feedback for use in their day-to-day anomaly surveillance system.
- 2021–2024 **PhD student**, Mines Paris - PSL, Geostatistics Unit
- Generative models and simulation-based inference for solving inverse problems, with applications to geology (GANs and Diffusion Models).
 - Proposed GAN inversion method for conditional generation using Gaussian Mixture Models.
 - Designed a novel discrete diffusion model for categorical spatial data using Markov Jump Chains.
 - Lectured on generative deep learning at ENS and Mines Paris.
- Feb–Aug 2021 **Research Intern**, Mines Paris - PSL, Geostatistics Unit
- Deep Learning alternatives to classical geostatistical simulation methods.
- Sep–Dec 2019 **Software Engineering Intern**, French Military
- Software development for a machine learning pipeline to extract structured data from unstructured military reports.

Education

- 2021–2024 **PhD in Applied Mathematics**, PSL Research University
- Geostatistics and Applied Probability doctorate. Advisors: Thomas Romary, Nicolas Dessasis
 - Subject: Deep Generative Models for Conditional Spatial Simulation ([link](#)).
- 2016–2021 **Computer Science & Software Engineering Degree**, EPITA
- French Engineering Degree - Equivalent to Master Degree.
 - Specialization in Cognitive Science and Artificial Intelligence (SCIA).

Certifications

- 2026 **TryHackMe - AI Security Course** Practical foundation in AI security. Learning how AI systems are built, where they break, and how to defend them.
- 2025 **Blue Dot - Technical AI Safety Course** Technical course to learn how to drive AI safety research and governance solutions.

Awards & Distinctions

- 2022 **GenHack 2022 (BNP Paribas & École Polytechnique)** - 1st place out of 64 international teams. Generative modeling for ocean temperature simulation.

Relevant Projects

- 2026 How LLMs Encode Grid Goals: Evidence for Agent-Centric Breadth-First Search - exploratory mechanistic analysis of spatial goals representations in LLMs. ([link](#))

Publications

- 2026 Transformer-based Diffusion models for Hydrological Time Series Probabilistic Imputation and Forecasting- *Arxiv preprint*, Bhavsar et al.
- 2024 A stable deep adversarial learning approach for geological facies generation - *Computers & Geosciences*, Bhavsar et al.

Skills

Programming Python (expert), C, C++ (advanced), CUDA, C# (basic)

Machine Learning PyTorch, HuggingFace Transformers, TransformerLens, TensorFlow/Keras, scikit-learn

Tools Git, Docker, AWS, SQL, Spark, Elasticsearch, LaTeX

Languages French (native), English (C1, TOEIC 990)

Personal

– French Boxing

2016–2020 Volunteer First Responder (PSE1 & PSE2 – French First-Aid certifications)

2021–2022 Boy Scout Leader - led teams, organized outdoor education, managed logistics & safety.